NAMANGA SECONDARY SCHOOL
FORM FOUR – TEST 1
INFORMATION AND COMPUTER STUDIES (ICS) EXAM

Time: 2 Hours

DATABASE MANAGEMENT SYSTEM (ACCESS)
PRACTICAL TEST

Name: _____ Stream: _____ Marks: _____ / 30

Instructions

1. This paper consists of one (1) question.
2. Answer all parts of the question.
3. This paper contributes to your Continuous Assessment (CA) for ICS. Your performance will be used, among other tests, to determine the best student in ICS in Form Four this year.
4. Show all the steps you use to arrive at your answer.
5. Cellular phones and any unauthorised materials are not allowed in the examination room.
6. Write your name and stream on every page of your answer booklet(s).

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1. SAFARI AUTO SPARES LTD is a company dealing with the sale of vehicle spare parts to garages across Tanzania. The company buys parts from registered suppliers and sells them to garages that place orders. All records are kept in three related tables named SUPPLIERS, PARTS and ORDERS. The field with an asterisk (*) in each table is the primary key for that table. Study the three tables below carefully and answer the questions that follow:

Table 1: SUPPLIERS

SupplierID*	SupplierName	Region	PhoneNumber	ContractStartDate
S01	Kilimanjaro Motors	Arusha	0754-112233	12/03/2019
S02	Bahari Auto Parts	Dar es Salaam	0715-556677	05/07/2020
S03	Nyanza Spares Co.	Mwanza	0688-990011	23/11/2018
S04	Ruaha Engineering	Iringa	0765-334455	30/01/2021
S05	Twiga Auto Supply	Arusha	0622-778899	14/06/2020
S06	Serengeti Motors	Mwanza	0713-221144	09/09/2017
S07	Coastal Spares Ltd	Tanga	0655-887766	18/04/2022

Table 2: PARTS

PartCode*	PartName	Category	UnitPrice (Tsh)	QtyIn Stock	SupplierID	DateReceived

P101	Brake Pad	Suspension	45000	8	S01	02/01/2023
P102	Alternator	Electrical	120000	5	S02	15/01/2023
P103	Bumper Set	Body	95000	14	S03	20/02/2023
P104	Radiator	Engine	150000	6	S04	03/03/2023
P105	Battery Terminal	Electrical	8000	40	S05	11/03/2023
P106	Ball Joint	Suspension	38000	9	S06	25/03/2023
P107	Bonnet Cover	Body	175000	3	S07	02/04/2023
P108	Fuel Injector	Engine	210000	4	S01	10/04/2023
P109	Shock Absorber	Suspension	67000	11	S02	18/04/2023
P110	Headlamp Cover	Electrical	26000	17	S03	05/05/2023
P111	Engine Mounting	Engine	54000	7	S05	12/05/2023
P112	Bonnet Lock	Body	15000	22	S06	30/05/2023

Table 3: ORDERS

OrderID*	PartCode	GarageName	OrderDate	QtyOrdered	DeliveryStatus
O1001	P101	Neema Garage	05/01/2023	10	Delivered
O1002	P104	Uhuru Auto Works	08/03/2023	3	Pending
O1003	P108	Furaha Motors	15/04/2023	2	Pending
O1004	P110	Baraka Garage	20/05/2023	15	Delivered
O1005	P101	Uhuru Auto Works	10/02/2023	5	Cancelled
O1006	P106	Neema Garage	02/04/2023	8	Delivered
O1007	P104	Baraka Garage	25/03/2023	4	Pending
O1008	P112	Furaha Motors	01/06/2023	20	Delivered
O1009	P108	Neema Garage	18/04/2023	1	Pending
O1010	P101	Furaha Motors	22/01/2023	6	Delivered

(a) Using a database management system program of your choice, create a database called SAFARI AUTO SPARES. Within it, create the three tables SUPPLIERS, PARTS and ORDERS exactly as shown above. Choose a proper and appropriate data type for every field, including all three primary key fields marked with an asterisk (*). Enter all the records shown in the three tables.

(b) Set the correct primary key for each of the three tables. In the PARTS table, SupplierID is a foreign key that links to the SUPPLIERS table. In the ORDERS table, PartCode is a foreign key that links to the PARTS table.

(c) Create the following two relationships and enforce referential integrity on both of them:

(i) A one-to-many relationship between the SUPPLIERS table and the PARTS table, using SupplierID as the common field.

(ii) A one-to-many relationship between the PARTS table and the ORDERS table, using PartCode as the common field.

(d) Apply the following controls to the fields listed below:

(i) Use the Lookup Wizard to set up the DeliveryStatus field in the ORDERS table so that, instead of typing, a user selects the value from a drop-down list. The drop-down list must contain only these three values: Pending, Delivered and Cancelled.

(ii) Validate the QuantityInStock field in the PARTS table so that negative values cannot be entered. Display the message "Quantity cannot be negative" if a user enters a value less than zero.

(iii) Validate the Category field in the PARTS table so that only the values "Engine", "Body", "Electrical" or "Suspension" are accepted. Display the message "Invalid category" for any other entry.

(e) Create the following queries:

(i) A query that displays all fields of the PARTS table for parts whose UnitPrice is greater than Tsh 50,000 AND whose QuantityInStock is less than 10. Save the query as "Critical Stock".

(ii) A query that displays all fields of the PARTS table for parts whose name begins with the letter "B" OR ends with the letter "r", restricted to parts in the "Engine" or "Electrical" category only. Save the query as "BE Parts".

(iii) A query, based on the PARTS table, that calculates a field called TotalValue (UnitPrice multiplied by QuantityInStock) and a second calculated field called VAT which is 18% of TotalValue. All fields of the PARTS table must be displayed alongside the two calculated fields. Save the query as "Stock Valuation".

(iv) A query that combines the SUPPLIERS and PARTS tables to display SupplierName, Region, PartName and UnitPrice for parts received between 01/03/2023 and 30/04/2023 inclusive. Save the query as "Supply Records".

(v) A query, based on the PARTS table, that displays parts whose UnitPrice is BETWEEN Tsh 20,000 and Tsh 100,000, and whose Category is NOT "Body". Save the query as "Mid Range Parts".

(vi) A query that combines all three tables (SUPPLIERS, PARTS and ORDERS) to display GarageName, PartName, Region and OrderDate for orders whose DeliveryStatus is "Pending" and whose part was supplied by a supplier from the "Arusha" region. Save the query as "Pending Arusha Orders".

(vii) A totals query, based on the ORDERS table, that groups records by PartCode and shows the sum of QtyOrdered for each part. The query must display only the parts that have been ordered more than once. Save the query as "Popular Parts".

(f) Create a form for each of the three tables using the Form Wizard. Name the forms "SupplierForm", "PartForm" and "OrderForm" respectively. Use the forms created to confirm that all records have been entered correctly.

(g) Using the PARTS table, create a report that groups records by Category and displays the PartName, UnitPrice and QuantityInStock for each part, together with the sum of TotalValue (UnitPrice × QuantityInStock) for each category group. Insert the current date as a header on the report. Save the report as "Category Report", then print all your work, including the tables, relationships, queries, forms and report created above.

(30 marks)